

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2022-23

Paper No. CHB-04A: Ancillary Chemistry-II (Basic Aspects of Chemistry)

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) "Petroleum and natural gas are stored forms of solar energy." Justify this statement.
- (b) Define Kranz anatomy. Draw a neat labeled structural representation of Kranz anatomy in a leaf.
- (c) Discuss the different factors that affect the rate of photosynthesis in plants.
- (d) Write down the chemical structures of Nylon-6,6 and Bakelite.
- (e) Differentiate between a cofactor and a prosthetic group.

Question 2.

- (a) What is a fuel cell? Give the electrode reactions of a hydrogen-oxygen fuel cell. [5]
- (b) Discuss intermolecular and intramolecular hydrogen bonds with one example of each. [5]
- (c) What is corrosion and what are its classifications? Discuss the mechanism of rust formation on iron. How can it be prevented by cathodic protection? [4]

Question 3.

- (a) Differentiate between isothermal and adiabatic processes. How do changes in enthalpy (ΔH), entropy (ΔS), and Gibbs free energy (ΔG) affect the spontaneity of a thermodynamic process? [5]
- (b) Give two examples each of extensive and intensive thermodynamic properties. [4]
- (c) What is a protein? State the significance of the primary and secondary structures of a protein. [5]

Question 4.

- (a) Define the Respiratory Quotient (RQ). Why is the RQ value for a carbohydrate substrate unity, while for substrates like fats and proteins it is less than one? ... [5]
- (b) What is an enzyme? Discuss the types and roles of enzymes in living things. [4]

- (c) Explain the Born-Haber cycle for the formation of ionic NaCl. What are the factors affecting the lattice energy of ionic solids? Among NaCl and MgO, which has higher lattice energy and why? [5]
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Question 5.

- (a) What are the different types of thermodynamic systems? Explain them briefly with suitable examples. [6]
- (b) How is the standard electrode potential (E°) different from the single electrode potential (E)? Write the Nernst equation for a single electrode. [3]
- (c) What is petroleum refining? Why is it carried out? Name the different fractions obtained during refining of petroleum along with their boiling ranges and carbon atom ranges. [5]
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Question 6.

- (a) What do you mean by the vulcanization of rubber? Write down the preparation method, properties, and applications of Polyneoprene and Buna-S rubber. [5]
- (b) What is the significance of the octane number and cetane number in petrol and diesel engines? Draw the structures of the organic compounds having the highest and lowest values of octane number. [5]
- (c) Define aerobic and anaerobic respiration with suitable chemical equations. [4]
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Question 7.

- (a) Describe the biological role and structural differences between starch and cellulose. [4]
- (b) Describe fibrous and globular proteins and their functions. [4]
- (c) Differentiate between the following pairs in brief: [4 × 1.5 = 6]
- (i) Respiration and Photorespiration
 - (ii) Coordinate covalent bond and Covalent bond
 - (iii) Photosystem I and Photosystem II in photosynthesis
 - (iv) Extensive properties and Intensive properties
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